

Evaluating AI-Generated Consumer Guidance Across Six Applied Domains

A Reference Implementation of the Universal Core Framework v1.0

Registered Protocol — Full Implementation Study

Owen Walcher

Independent Researcher · Grand Junction, CO USA

Correspondence: owalcher@gmail.com · ORCID: 0009-0000-4716-2267

Companion to: Walcher (2026a), Universal Core Framework v1.0. This paper inherits all methodology from that framework. Date: May 2026

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Abstract

Large language models (LLMs) are increasingly consulted for high-stakes consumer decisions, yet general benchmarks do not capture the consumer-centric failure modes that matter in these settings. As argued in the companion framework paper (Walcher 2026a), evaluating such guidance requires a methodology built for procedural, jurisdictional, and safety-sensitive judgment. This protocol instantiates that methodology — the Universal Core Framework v1.0 — across six empirical domains: legal literacy, insurance navigation, home buying, consumer protection, home remodeling, and health advocacy.

Classified as a Full Implementation Study (Walcher 2026a, §3.2.4), the design comprises three phases: development of a calibrated library of exactly 270 prompts (15 per complexity tier × 3 tiers × 6 domains); standardized collection of responses from five leading LLMs under disclosed conditions; and dual-track evaluation combining domain-expert review with an automated and LLM-as-judge pipeline, calibrated in two pilot domains. Responses are scored on the framework's six dimensions using its anchored five-point scale; behavioral signatures are coded systematically per the framework. The study supports inferential conclusions and effect-size interpretation within the framework's claim-strength governance (Walcher 2026a, §15), and will report domain-specific performance scorecards, human–automated concordance, a consumer-AI error taxonomy, behavioral-signature patterns, and a reusable prompt library, together with evidence for or against five pre-registered hypotheses.

Keywords: AI evaluation; consumer guidance; registered protocol; reference implementation; behavioral signatures; applied AI

1. Introduction

1.1 The Consumer Guidance Gap

Consumers increasingly consult LLMs for guidance on legal rights, insurance claims, home purchases, consumer disputes, medical billing, and home remodeling — domains where inaccurate, incomplete, or insufficiently caveated advice can cause financial loss, legal jeopardy, or physical harm. As specified in the companion framework (Walcher 2026a, §1), general-purpose benchmarks such as MMLU, HumanEval, and GSM8K measure academic and technical capability and do not capture these consumer-centric failure modes; that argument is not repeated here. What distinguishes the present study is the six concrete domains in which the framework is instantiated. Each satisfies the framework’s domain-eligibility criteria (Walcher 2026a, §7) — material consequences, jurisdictional variability, procedural complexity, a history of expert mediation, and high variance across authoritative sources — and together they span the regulatory regimes (state landlord-tenant law; the McCarran-Ferguson Act 1945; RESPA 1974; the International Residential Code 2021; HIPAA 1996; the ACA 2010; the No Surprises Act 2020) under which consumer guidance must be correct to be safe.

1.2 Relationship to the Universal Core Framework

This study is the worked Reference Implementation of the Universal Core Framework v1.0 (Walcher 2026a). It adopts the Universal Core without modification for all immutable components: the six evaluation dimensions and their definitions, the anchored five-point scoring scale, the prompt complexity tiers and framing taxonomy, the dual-track scoring procedures, the behavioral-signature constructs and coding rules, and the governance rules for claims, contextual disclosure, deviation accounting, reliability escalation, and reproducibility. None of these are re-derived in this paper; each is cited to its location in the framework. The present paper specifies only the adaptable, implementation-specific layer (Walcher 2026a, §2.3): the choice of six domains, the prompt library and its exact sampling plan, the six-model set, the pilot-domain choice, the statistical analysis plan, and five pre-registered hypotheses. Section 3.1 provides a component-by-component mapping (Table 3.1) so that no implementation choice can be mistaken for a universal rule.

Study Classification Declaration (Walcher 2026a, §3.2.4, Table 3.1)

Classification: Full Implementation Study. This study meets all stated minima: 30–45 prompts per domain (here, exactly 45); all three complexity tiers; formal reliability analysis; calibration documentation; adjudication documentation; and full reproducibility materials.

Permitted claims: inferential conclusions; effect-size interpretation; longitudinal comparison; cross-domain synthesis — each bounded by the study’s realized reliability (Walcher 2026a, §11.4).

Prohibited claims: any inference about internal model intent, alignment mechanisms, training dynamics, or optimization objectives absent causal experimental methodology (Walcher 2026a, §15.3).

1.3 Institutional Context and Positionality

The study is conducted within an independent consumer-education and publishing program with documented experience of the questions consumers ask across several of these domains — experience that informs the prompt library. To mitigate the associated bias, the study relies on pre-registration, the framework’s standardized rubric, evaluator blinding where feasible, and transparent documentation, so the design reflects a reproducible protocol rather than editorial judgment. The competing interest this creates is declared in full in the Declarations.

1.4 Research Questions

- RQ1: How accurate, complete, and actionable is AI-generated guidance across the six domains when evaluated with the framework rubric against expert benchmarks?
- RQ2: Do performance patterns vary systematically by domain, and which domains show the greatest strengths and vulnerabilities across dimensions?
- RQ3: Can the standardized prompt-response-evaluation pipeline produce reproducible, publication-quality findings across heterogeneous consumer domains?

1.5 Study Objectives

The study (1) instantiates the framework with a calibrated prompt library and a documented five-model pipeline; (2) generates domain-specific quality scores comparable across domains, tiers, and models; (3) identifies systematic strengths and weaknesses, with attention to jurisdiction sensitivity, safety caveats, and judgment-dependent reasoning; and (4) produces a replicable implementation extensible to further domains and models.

2. Related Work in Consumer-Domain AI Evaluation

The general critique of capability benchmarks is made in the framework paper (Walcher 2026a, §1–§2) and is not restated here; this section is limited to consumer-specific AI behavior within the six study domains. In law, evaluations find strong factual legal knowledge (Katz et al. 2024) but weaker jurisdiction-specific application and multi-step reasoning, with most work targeting professional rather than consumer-level questions. In insurance, studies report difficulty explaining policy exclusions and state-specific appeal procedures under each state’s Insurance Code (McCarran-Ferguson Act 1945). In real estate, evaluations concentrate on automated valuation models rather than the consumer guidance central to home buying under RESPA (RESPA 1974). In consumer protection, formal evaluation of guidance on warranty enforcement (Magnuson-Moss Warranty Act 1975) or complaint filing with agencies such as the CFPB (CFPB 2011) is limited. In home remodeling, evaluation is nearly absent, with informal assessments noting omitted safety warnings and missing code references (International Code Council 2021). In health advocacy, substantial work evaluates AI health information (Wang et al. 2024; Sandmann et al. 2024; Singhal et al. 2023), but largely on clinical accuracy rather than the advocacy and navigation tasks patients face as consumers. The consistent finding across these domains — strong factual recall, weak jurisdictional nuance, weak multi-step procedure, and inconsistent safety caveats — motivates a unified, cross-domain instrument, which this study supplies by deploying the framework across all six at once.

3. Methods

This study inherits its methodology in full from the Universal Core Framework v1.0 and specifies only implementation choices. The subsections below describe domains, models, the prompt sampling plan, the application of the framework rubric, behavioral-signature coding, the collection pipeline, sample size, and consumer-impact interpretation. Definitions of the dimensions, scale, tiers, framing taxonomy, scoring procedures, and governance rules are not reproduced; they are cited to the framework.

Note on scope. Everything in this section is an implementation choice (Walcher 2026a, §2.3, "adaptable components"). Nothing here redefines, narrows, or extends an immutable component of the Universal Core; where this paper appears to describe a rule, it is applying a universal rule to this study, not creating one.

3.1 Study Design, Classification, and Inheritance Mapping

The study is a pre-registered systematic evaluation using a prompt-response-evaluation pipeline, classified as a Full Implementation Study (declared in §1.2). It comprises three sequential phases: (1) prompt library development; (2) standardized multi-model response collection; and (3) dual-track evaluation merged into a unified analysis dataset. Table 3.1 maps each universal component to its treatment in this implementation, so reviewers can confirm that all immutable components are inherited unchanged and locate every adaptation.

Universal Core component	Treatment here	Where specified
Six dimensions + definitions	Inherited unchanged	Walcher 2026a, App. A
Five-point scoring scale	Inherited unchanged	Walcher 2026a, App. B
Complexity tiers + framing taxonomy	Inherited unchanged	Walcher 2026a, §8, App. D
Dual-track scoring procedures	Inherited unchanged	Walcher 2026a, §11
Behavioral-signature constructs + coding rules	Inherited unchanged	Walcher 2026a, §6, App. C
Claim-strength, contextual disclosure, deviation, reliability, versioning	Inherited unchanged	Walcher 2026a, §9, §11.4, §13, §15, §17, §18
Domain selection	Adapted (6 domains chosen)	This paper, §3.2
Prompt library + sampling	Adapted (N = 270; 15/tier/domain)	This paper, §3.3, App. B
Model set	Adapted (6 models)	This paper, §3.4
Pilot-domain choice	Adapted (Legal, Health)	This paper, §3.5
Statistical analysis plan	Adapted (within §14)	This paper, §5

Universal Core component	Treatment here	Where specified
	governance)	
Hypotheses	Adapted (H1–H5)	This paper, §4
Consumer-impact interpretation	Added (study-specific)	This paper, §3.10

Table 3.1. Universal → Implementation mapping. Immutable components are inherited unchanged; adaptations are confined to the rows marked "Adapted/Added."

3.2 Domain Specification

The six domains and their representative sub-domains are listed in Table 3.2. Each was selected for satisfying the framework’s eligibility criteria (Walcher 2026a, §7); the general rationale is given there and not repeated.

Domain	Representative sub-domains
Legal Literacy	Family, estate, and elder law; personal injury and consumer protection; criminal, civil-rights, and traffic law; business, employment, and tax law; real estate and landlord-tenant; self-advocacy
Insurance Navigation	Policy anatomy and coverage gaps; health-plan comparison; auto and property insurance; life and disability planning; claims filing and denial appeals; bad-faith practices
Home Buying	Transaction stages; agency and fiduciary duties; mortgage qualification; offer negotiation and contract terms; escrow, inspections, and closing; predatory-practice protections
Consumer Protection	Identity-theft recovery; financial and investment fraud; digital and cyber fraud; marketplace and service fraud; government and institutional fraud; fraud psychology
Home Remodeling	Scope and cost estimation; contractor vetting and disputes; permits, codes, and DIY safety thresholds; plumbing, electrical, and HVAC; structural and trade-specific guidance; energy efficiency
Health Advocacy	Patient rights, consent, and HIPAA; pre-authorization and appeals; medical-billing disputes; discharge planning and care transitions; health equity and digital health; preventive care

Table 3.2. Evaluation domains and sub-domains.

3.3 Prompt Library and Sampling Plan

The library comprises exactly 270 unique prompts: within each of the 6 domains, exactly 15 prompts are written at each of the 3 complexity tiers (Tier 1 factual, Tier 2 procedural, Tier 3 judgment-dependent; Walcher 2026a, §8.1), yielding 45 prompts per domain and an

uncompromised total of $N = 270$. This fixed allocation ($15 \times 3 \times 6$) is balanced by design, eliminating cell-size imbalance in the Domain $\times$ Tier analysis. Every prompt records its complexity tier and its framing category from the framework’s taxonomy (neutral, emotionally expressive, adversarial, expertise-signaling, authority-primed, or vulnerable-user; Walcher 2026a, §8.2); framing is a required field for every prompt in the published library, without exception, so that framing can be held constant or modeled as a factor. Prompts use a standardized format pairing a domain term in quotation marks with an action verb to produce consumer-natural questions, and all are reviewed by at least one domain expert for clarity, representativeness, and difficulty calibration before formal collection.

3.4 Model Selection and Diversity Justification

Responses are collected from five LLMs satisfying the framework’s model-selection minima (Walcher 2026a, §10: at least three meaningfully distinct models, at least one from a recognized reference ecosystem). The set deliberately spans four sources of architectural and behavioral diversity: (1) frontier instruction-tuned model from distinct alignment lineage — a GPT-4-class model (OpenAI 2023); (2) a separate large-scale multimodal lineage — a Gemini-class model; (3) a model emphasizing real-time data integration and divergent response tendencies — a Grok-class model; and (4) two retrieval-augmented (RAG) architectures that ground answers in live search — a Perplexity-class system and Microsoft Copilot (an enterprise-integrated GPT system). This composition exceeds the three-model minimum and varies along the dimensions the framework treats as material — base architecture, alignment approach, and retrieval strategy (non-RAG vs. RAG) — making it sufficiently diverse to support cross-model comparison rather than convenience sampling. Each prompt is submitted with standardized consumer-seeking system instructions; temperature is fixed at 0.3, the maximum token limit permits full responses, and a single response is collected per prompt-model pair. Each response is captured in raw text, including model identifier and version, system prompt, and submission timestamp.

3.5 Application of the Framework Rubric

Responses are evaluated through the framework’s dual-track procedure (Walcher 2026a, §11). The six dimensions and their anchored five-point descriptors are applied exactly as defined in the framework (Walcher 2026a, Appendices A–B) and are not reworded here, to avoid drift from the immutable definitions; Table 3.3 is a one-line index only. Critically, each of the six dimensions is scored independently for every response — raters assign each dimension on its own evidence rather than forming a single overall impression — to prevent the halo effects the framework’s independent-scoring requirement is designed to avoid (Walcher 2026a, §2.2, §5.1).

Dimension	Type	One-line index (full descriptors: Walcher 2026a, App. B)
Accuracy	Descriptive	Factual correctness; absence of hallucination

Dimension	Type	One-line index (full descriptors: Walcher 2026a, App. B)
Completeness	Descriptive	Coverage of components needed to act
Actionability	Normative	Clarity and sequencing of next steps
Safety	Normative	Appropriate caveats; avoidance of harm
Jurisdiction Sensitivity	Normative	Awareness of regional/jurisdictional variation
Transparency	Descriptive	Disclosure of limits, uncertainty, and refusals

Table 3.3. Index to the six framework dimensions. See Walcher 2026a, Appendix B for the full anchored descriptors, which govern.

3.5.1 Track A — Expert Panel Review and Pilot Domains

Expert review follows a phased design centered on two pilot domains: Legal Literacy and Health Advocacy. These two are chosen not only because they are high-risk but because they are the domains best suited to validating the automated track before it is relied upon elsewhere: both turn heavily on the normative, safety-sensitive dimensions (Safety, Jurisdiction Sensitivity, Actionability) where automated scoring is most likely to diverge from expert judgment, and both have clear professional reference standards (licensed attorneys/paralegals; patient advocates/healthcare administrators) against which expert anchors can be set. Establishing human–automated concordance in precisely these hardest cases provides the strongest justification for extending automated scoring to the remaining four domains. One to two domain experts per pilot domain (2–4 total) score a representative subset using the framework rubric; inter-rater reliability is assessed with Krippendorff’s alpha (Krippendorff 2018), target $\alpha \geq 0.70$ per dimension, with escalation handled per the framework (Walcher 2026a, §11.4, App. K). The remaining four domains are then evaluated primarily through the automated track with targeted spot-check expert review.

3.5.2 Track B — Automated Scoring Pipeline

The automated track combines structural/checklist scoring (programmatic detection of caveats, step-by-step structure, jurisdictional acknowledgment, conditional language, and actionable next steps against domain checklists) with LLM-as-judge evaluation (Zheng et al. 2023), in which a separate judge model returns dimension-level scores with justifications. The judge model is disclosed in full per the framework’s AI-rater transparency rule (Walcher 2026a, §11.3, App. G); the completed disclosure for this study is Appendix D.2. Reference-answer calibration in the two

pilot domains develops expert-consensus outlines used to calibrate the automated approaches, with a target composite concordance of Pearson $r \geq 0.70$ before extension.

3.6 Behavioral Signature Coding

Behavioral signatures are treated as a core measurement in this study, not an afterthought. Every collected response is coded for the framework’s behavioral signatures — tone mirroring, persona overfitting, overconfidence, actionability inflation, procedural drift, hallucinated institutional facts, ambiguity amplification, and uncertainty suppression — using the operational definitions, positive/negative indicators, and evidence-citation requirement of the Behavioral Signature Coding Manual (Walcher 2026a, §6, Appendix C). Per those rules, each coded instance is tied to a quoted span of model output with a documented rationale, and no causal interpretation is attached. Signature prevalence is reported as frequencies and rates broken down by domain, tier, framing category, and model (see §5.5), so that behavioral tendencies — for example, whether overconfidence rises with tier or whether tone mirroring tracks emotionally expressive framing — can be examined systematically rather than anecdotally.

3.7 Data Pipeline Architecture

The pipeline has five auditable stages: (1) prompt ingestion into a version-controlled repository with unique identifiers (e.g., LEGAL-T2-007); (2) automated API collection with full request/response logging; (3) evaluation ingestion (expert scores via a standardized form; automated scores via scripts); (4) merging into a unified dataset keyed by prompt ID, model ID, domain, and tier, including all individual scores, reliability statistics, and behavioral codes; and (5) quality assurance detecting missing or out-of-range data, computing reliability per domain and dimension, and flagging zero-variance prompts. Response failures (refusals, truncation, API failure, timeout, malformed output, content-filter interruption) are categorized and retries governed exactly per the framework (Walcher 2026a, §12); no failed response is silently dropped.

3.8 Temporal Collection and Mid-Window Disruption Protocol

Collection timing and interaction context are disclosed per the framework (Walcher 2026a, §9, §13; completed disclosures in Appendices D–E). Collection is batched by domain block. Because this is an inferential study, the study adopts an explicit mid-window disruption protocol to safeguard temporal consistency: if an API provider pushes a major model version change while a domain block is actively being collected, all previously collected responses for that model within the active domain block are discarded and re-collected within a compressed 48-hour window under the new version, and the event is logged in the model-drift documentation (Appendix E) and the deviation table (Appendix C). Minor or interpretive provider changes that do not alter model version are recorded but do not trigger re-collection. The temperature setting (fixed at 0.3) and retrieval-augmentation status of each model are recorded as part of the contextual-state disclosure (Appendix D.1), since both materially affect reproducibility.

3.9 Sample Size and Power Analysis

With $N = 270$ prompts across 6 domains, 3 tiers, and 6 models, the study generates 1,620 evaluation instances for automated scoring; the two-domain expert validation phase yields 540 paired expert-automated scores. For the two-way ANOVA (Domain $\times$ Tier) on each dimension, detecting a medium effect (Cohen's $f = 0.25$) at 80% power and $\alpha = 0.05$ requires roughly 64 observations per cell across 18 cells ($\sim 1,152$ observations). With exactly 15 prompts per tier per domain and six models, the design provides 90 observations per cell, comfortably exceeding the threshold; for larger effects ($f = 0.40$) the margin is greater. Inferential analyses account for the nested data structure (prompts within domains) per the framework's statistical governance (Walcher 2026a, §14.1).

3.10 Consumer Impact Interpretation Brackets

To connect the framework's numeric outputs to real-world consumer outcomes — interpretation the framework deliberately leaves to implementations to keep itself domain-agnostic — this study defines the following cumulative-score brackets on the 6–30 composite. These brackets are study-specific interpretive aids and are reported alongside, not in place of, the per-dimension scores.

Cumulative score	Bracket	Consumer-impact interpretation
< 18	Deficient Guidance	Unsafe or structurally incomplete; requires comprehensive professional rewriting before any consumer deployment.
18–25	Conditional Guidance	Factually acceptable but lacking critical actionability or structural transparency; usable only with targeted expert revision.
26–30	Production-Ready Guidance	Suitable for direct public consumer-advocacy distribution with minimal editorial oversight.

Table 3.4. Consumer Impact Interpretation Brackets (study-specific; applied to the 6–30 composite).

3.11 Ethical Considerations

The study does not involve human subjects as defined under 45 CFR 46 (U.S. Department of Health and Human Services 2018); expert panel members participate as professional consultants and provide informed consent, so IRB approval is not required. The study commits to publishing results regardless of whether they confirm its hypotheses and adheres to open-science principles (Chambers & Tzavella 2022): the protocol, prompt library, analysis code, and

anonymized data will be released publicly, with the rubric and instruments available via the companion framework paper.

4. Hypotheses

The five hypotheses below are pre-registered as confirmatory analyses under the framework's confirmatory/exploratory distinction (Walcher 2026a, §4): each is specified before data collection with a predefined test, and each is reported as confirmatory regardless of outcome. Any analysis not listed here is exploratory (§5.6).

- H1 (Tier effect on Accuracy and Completeness): guidance will score higher on Accuracy and Completeness for Tier 1 than Tier 3 across all six domains, with a medium-to-large effect (Cohen's $d \geq 0.5$).
- H2 (Domain effects on overall performance): Health Advocacy and Legal Literacy will yield the lowest composite scores, reflecting dense, frequently updated regulatory frameworks (HIPAA 1996; ACA 2010; state landlord-tenant and consumer-fraud statutes).
- H3 (Domain effects on Safety): Safety scores will be lowest in Home Remodeling and Health Advocacy, reflecting heightened physical-harm risk from uncaveated advice. (H2 concerns composite scores; H3 concerns the Safety dimension.)
- H4 (Inter-model variability by tier): score variability across models for a prompt will be greater for Tier 3 than Tier 1, as judgment-dependent scenarios permit a wider range of defensible responses.
- H5 (Concordance by dimension): human–automated concordance will be highest for Accuracy and Completeness, moderate for Transparency, and lowest for Actionability, Safety, and Jurisdiction Sensitivity, reflecting the normative dimensions' greater subjectivity (Walcher 2026a, §5.2).

5. Analysis Plan

All analyses align with the Full Implementation classification and the framework's statistical and claim-strength governance (Walcher 2026a, §14–§15). Confirmatory analyses (§5.2) test H1–H5; exploratory analyses (§5.6) are reported as such.

5.1 Descriptive Statistics

For each dimension, the study reports mean, standard deviation, median, and interquartile range by domain, tier, model, and their two-way cross-tabulations, plus the cumulative composite and its bracket distribution (§3.10). Distributions are checked for floor and ceiling effects. Visualizations include box plots by domain within each dimension, Domain × Model and Domain × Tier heat maps, and per-model radar charts.

5.2 Inferential Analyses (Confirmatory)

Within the framework's claim-strength governance, the study supports inferential conclusions and effect-size interpretation (Walcher 2026a, §15, Table 15.1). Two-way ANOVA (Domain × Tier) is run per dimension with model as a random factor; significant omnibus effects are

followed by Tukey HSD with family-wise error at $\alpha = 0.05$. Mixed-effects regression models scores as a function of Domain, Tier, Model, and two-way interactions, with Prompt nested within Domain as a random effect (Walcher 2026a, §14.1). Effect sizes are reported as Cohen's d for pairwise comparisons and partial eta-squared for ANOVA effects. Reported claim strength is conditioned on realized reliability: dimensions falling below the robust threshold are reported with reduced inferential confidence per the reliability-escalation rules (Walcher 2026a, §11.4, App. K).

5.3 Concordance Analysis

Human–automated concordance (testing H5) is assessed via Pearson and Spearman correlations, Bland-Altman plots for systematic bias, Cohen's kappa for pass/fail agreement (pass = ≥ 3.0 on all dimensions), and intraclass correlation coefficients for absolute agreement between tracks.

5.4 Cross-Domain Synthesis

Domain-level effect sizes are pooled using random-effects synthesis (Borenstein et al. 2021; Viechtbauer 2010): random-effects models pool standardized mean differences across models within each domain; forest plots display domain estimates with 95% CIs; heterogeneity is assessed via I^2 and Cochran's Q ; sensitivity analyses exclude individual models or domains; and moderator analyses test whether tier, architecture, or domain characteristics explain heterogeneity. Cross-domain synthesis is permitted under the Full Implementation classification.

5.5 Behavioral Signature Reporting

Behavioral signatures coded per §3.6 are reported quantitatively: prevalence (counts and rates) for each signature, cross-tabulated by domain, tier, framing category, and model, with representative quoted exemplars. Where a study-classification permits, associations between framing category and signature prevalence (e.g., tone mirroring under emotionally expressive framing) are tested descriptively. Behavioral findings are presented as observational patterns consistent with the framework's descriptive stance (Walcher 2026a, §6.1) and carry no causal interpretation.

5.6 Exploratory Analyses

The following are explicitly designated exploratory and are reported separately from the confirmatory results, in keeping with the framework's required boundary (Walcher 2026a, §4.3): qualitative thematic analysis of expert free-text comments; development of a consumer-AI error taxonomy (factual error, omission, unsafe recommendation, jurisdictional error, outdated information, logical error, excessive hedging); any newly observed behavioral signatures beyond the eight catalogued; and prompt-difficulty calibration analysis comparing intended (expert-rated) with observed (performance-based) difficulty. No exploratory result is reported as a preregistered confirmation; any promising exploratory finding is framed as a hypothesis for a subsequent study.

5.7 Software and Reproducibility

Analyses use Python 3.10+ (pandas, NumPy, SciPy, statsmodels, scikit-learn, matplotlib/seaborn) and R 4.3+ (lme4, irr). All scripts, the prompt library, and the anonymized dataset are published in a public repository, meeting the framework’s study-specific reproducibility requirements for Full Implementation studies (Walcher 2026a, §16.2). That repository is established here: <https://github.com/owalcher/askafriend-AI-consumer-analysis>

6. Expected Outcomes

The study is expected to produce: (1) domain-specific performance scorecards across the six dimensions, stratified by tier and model, with composites mapped to the consumer-impact brackets; (2) human–automated concordance evidence by dimension; (3) a consumer-AI error taxonomy; (4) behavioral-signature prevalence patterns by domain, tier, framing, and model; (5) a reusable, calibrated 270-prompt library; and (6) evidence for or against each pre-registered hypothesis. Practically, these inform AI-assisted publishing workflows, consumer-education priorities, deployment best practices, and regulatory discussion relevant to FTC consumer-protection guidance (Federal Trade Commission 2023). Academically, the study is believed to be the first cross-domain consumer-AI evaluation under a shared, externally specified methodology and the worked Reference Implementation of the Universal Core Framework.

7. Limitations

Findings reflect a point-in-time snapshot of rapidly evolving models; the protocol mitigates this through version, parameter, and timestamp documentation and the mid-window disruption protocol (§3.8). No finite prompt set captures the full diversity of consumer scenarios, though the documented methodology supports library expansion. Concentrating full expert review in two pilot domains is a deliberate rigor-feasibility trade-off, and the $\alpha \geq 0.70$ target, appropriate for applied research, is below thresholds used for high-stakes assessment. Prompts primarily reflect U.S. federal and state contexts, limiting generalizability to other jurisdictions. Automated scoring may miss nuance perceptible to experts, which the concordance analysis is designed to quantify. Behavioral-signature coding remains partially interpretive despite operational rules. Finally, per the framework’s validation philosophy (Walcher 2026a, §1.3, §21), this single implementation contributes to but does not by itself establish the framework’s validated utility.

8. Conclusion

This protocol describes a pre-registered, cross-domain evaluation of AI-generated consumer guidance conducted as a Full Implementation of the Universal Core Framework v1.0. By inheriting its methodology from the framework and specifying only the implementation layer — six domains, a fixed 270-prompt library, a six-model set, two pilot domains, systematic behavioral-signature coding, and an analysis plan bounded by the framework’s claim-strength governance — the study demonstrates the framework in operation while keeping the methodology cleanly separable from this deployment. Consistent with the framework’s deviation-accounting requirement, all deviations from this plan will be documented not only in the protocol’s deviation table (Appendix C) but in the published results paper, using the

framework’s template (Walcher 2026a, App. F), so that the gap between plan and execution remains visible downstream. The pipeline is extensible to further domains and models, supporting a cumulative research program on a common methodological foundation.

Declarations

Funding

This study is internally funded as part of an independent program of consumer-facing AI evaluation research. No external funding was received.

Competing Interests

The author operates consumer-education resources in several of the domains under evaluation. This is declared as a potential non-financial competing interest. The same disclosure is reflected in the AI-assistance and author-contributions statements below; the language is intended to be consistent across all three. Mitigation includes pre-registration, the framework’s standardized rubric, blinding where feasible, and public release of all materials. The author declares no financial competing interests.

Ethics Approval and Consent to Participate

This study does not involve human subjects as defined under 45 CFR 46 (U.S. Department of Health and Human Services 2018). Expert panel members participate as professional consultants and provide informed consent prior to engagement; IRB approval is not required.

Data and Code Availability

Upon completion, the complete prompt library with difficulty-calibration metadata, the anonymized analysis dataset (CSV with data dictionary), and all analysis scripts (Python and R) will be released via a public repository. The evaluation rubric and scoring instruments are published with the companion framework paper (Walcher 2026a). A pre-registration record of the hypotheses and analysis plans has been filed prior to data collection. <https://github.com/owalcher/askafriend-AI-consumer-analysis>

Declaration of Generative AI and AI-Assisted Technologies

AI assistance was used for editorial and formatting support during manuscript preparation. Within the study’s evaluation pipeline, an AI system serves as the secondary rater (Track B), disclosed in full per the framework’s AI-rater transparency requirement (Walcher 2026a, §11.3; Appendix D.2). All substantive intellectual content — research design, framing, methodological decisions, hypotheses, analysis plan, and interpretation — was produced by the author, who takes full responsibility for the manuscript. No generative AI system contributed original scientific claims, data, or conclusions.

Author Contributions

The author solely conceived and designed the study, will collect data, oversee scoring and adjudication, conduct analysis, and wrote this manuscript, consistent with the competing-interest disclosure above. The work is part of a broader research program on the systematic evaluation of AI-generated consumer guidance.

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Appendices

These appendices contain implementation-specific materials only. Methodological definitions and reusable templates live in the companion framework paper (Walcher 2026a) and are cited, not reproduced. The Universal → Implementation mapping is given in Table 3.1.

Appendix A — Prompt Library (Implementation)

The Legal domain library is substantially complete and undergoing investigator hand-testing prior to formal collection; the remaining five domains are built with the same methodology. Hand-testing confirms that each prompt elicits substantively different responses across tiers, removes prompts that are ambiguous, jurisdiction-overloaded, or outside consumer utility, and produces the first published demonstration analyses. Before formal collection, each platform is asked to self-report its model version, which is recorded with the response metadata. The Legal library is organized into the modules in Table A1; the complete 270-prompt library (with tier, framing category, and difficulty metadata for every prompt) is released as Supplementary File S3. Representative prompts across three modules and all three tiers appear in Tables A2–A4.

Module	Subcategories covered
Business & Commercial	LLC formation, contracts, employment law, bankruptcy, real estate, landlord-tenant, tax
Family Law	Divorce, estate planning, elder law, adoption, guardianship
Injury & Civil Litigation	Personal injury, medical malpractice, workers' compensation, consumer protection, civil procedure
Criminal & Civil Rights	Criminal defense, DUI/traffic, civil rights
IP & Technology	Copyright, patent, trademark, cybersecurity, media law
Specialized Practice Areas	Immigration, military law, environmental law, education, aviation/maritime

Table A1. Legal domain prompt library modules.

Tier	Representative prompt (Injury & Civil Litigation)
1 — Factual	What is "Pain and Suffering"?
2 — Procedural	How do you use a "Daily Pain Journal" and medical testimony to quantify non-economic damages in a demand letter?
3 — Judgment	Should you take a "Settlement" now for a known amount, or go to "Trial" where a jury might award

Tier	Representative prompt (Injury & Civil Litigation)
	zero or triple that amount?

Table A2. Sample prompts — Injury & Civil Litigation.

Tier	Representative prompt (Business & Commercial)
1 — Factual	What is an "LLC"?
2 — Procedural	What are the steps to draft an "Operating Agreement" and file "Articles of Organization" with the Secretary of State?
3 — Judgment	Should you form a "C-Corp" to attract venture capital, or an "LLC" to simplify taxes and avoid double taxation?

Table A3. Sample prompts — Business & Commercial.

Tier	Representative prompt (Family Law)
1 — Factual	What is a "Power of Attorney" (POA)?
2 — Procedural	How do you legally invoke a "springing" Power of Attorney when a family member becomes incapacitated?
3 — Judgment	Is it better to grant a broad General POA to a single child, or a Limited POA to multiple children to prevent financial elder abuse?

Table A4. Sample prompts — Family Law.

Appendix B — Prompt Allocation (Locked: N = 270)

Final, locked sampling plan. Counts are fixed, not ranges; the framework permits 30–45 prompts per domain for Full Implementation studies and this implementation locks the maximum, exactly 45 per domain.

Domain	Tier 1	Tier 2	Tier 3	Domain total
Legal Literacy	15	15	15	45
Insurance Navigation	15	15	15	45
Home Buying	15	15	15	45

Domain	Tier 1	Tier 2	Tier 3	Domain total
Consumer Protection	15	15	15	45
Home Remodeling	15	15	15	45
Health Advocacy	15	15	15	45
Total	90	90	90	270

Table B1. Locked prompt allocation: 15 per tier $\times$ 3 tiers $\times$ 6 domains = 270 unique prompts.

Appendix C — Deviation Table

All deviations from this pre-registered protocol will be documented here using the exact template in Universal Core Appendix F (Walcher 2026a), and the same deviation table will be carried into the published results paper. No deviations exist at registration; the table is populated during execution. A mid-window model-version change (per §3.8) is logged here in addition to Appendix E.

Deviation	Rationale	Preregistered?	Likely impact
[None at registration]	—	—	—
[e.g., mid-window version change $\rightarrow$ block re-collected]	[Provider pushed major version update]	[No — handled per §3.8 protocol]	[Temporal consistency preserved; logged in App. E]

Table C1. Deviation table (second row illustrative; template per Walcher 2026a, App. F).

Appendix D — Contextual-State and AI-Rater Disclosure

Completed disclosures for this study, following the framework templates (Walcher 2026a, Appendices G–H).

D.1 Contextual-State Disclosure

Field	Disclosure for this study
Context classification	Semi-Controlled (fresh session per prompt)
Account persistence	Disabled where supported; no logged-in personalization
Thread isolation	Yes — each prompt submitted in a new thread
Prompt ordering	Fixed, recorded order; identical across models
Session continuity	Single-turn; no conversational memory

Field	Disclosure for this study
Temperature	Fixed at 0.3 for all models (recorded per response)
Model version(s)	Recorded per response (model self-report + API metadata)
Timestamps	Per-response submission timestamps logged
Retrieval augmentation	Enabled for RAG models (Perplexity-, Copilot-class); source recency logged; disabled/non-RAG status logged for the others

Table D1. Contextual-state disclosure for the six-domain study.

D.2 AI-Rater Disclosure

Field	Disclosure for this study
Judge-model identity	[Judge model family — fixed at registration]
Judge-model version	[Exact version / API string]
Access date(s)	[Scoring collection window]
Scoring prompt(s)	Verbatim rubric-scoring prompt published with S1
Scoring instructions	Framework rubric (Walcher 2026a, App. B) provided to judge, unmodified
Output format	Independent dimension-level 1–5 scores with brief justifications
Adjudication procedure	Human analyst adjudicates; human authority is final
Known limitations	Documented refusal/truncation handling per §12

Table D2. AI-rater disclosure for the six-domain study.

Appendix E — Model and Collection Documentation

Per the framework’s temporal collection governance (Walcher 2026a, §13, App. L) and the mid-window disruption protocol (§3.8).

Field	Disclosure for this study
Collection start / end	[Dates — fixed at registration]

Field	Disclosure for this study
Collection duration	[Total elapsed; batched by domain block]
Models and versions at start	GPT-4-class; Claude-class; Gemini-class; Grok-class; Perplexity-class; Copilot — exact versions logged
Version changes during window	[Logged; major change → block re-collected within 48 h per §3.8]
Retrieval freshness	[For RAG models: index/source recency]
Known platform updates	[Provider-announced changes near the window]
Re-collection actions	[Prompts re-collected after a change, with reason and timing]

Table E1. Model and collection documentation.

Appendix F — Planned Supplementary Materials (Study-Specific Deliverables)

The following are deliverables of this study, not of the framework. Framework templates and rubric instruments are released with Walcher (2026a).

- Supplementary File S1 — Automated Scoring Pipeline Technical Specification (this study): checklist rules, LLM-as-judge prompt templates, and validation procedures.
- Supplementary File S2 — Data Dictionary for the Analysis Dataset (this study): variable definitions, types, coding schemes, and metadata, including behavioral-signature codes.
- Supplementary File S3 — Complete 270-Prompt Library (this study): every prompt with domain, tier, framing category, and observed-difficulty metadata.
- Supplementary File S4 — Analysis Scripts and Reproducibility Package (this study): Python and R scripts for all analyses.

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